



Statistical Inference

FPM 2026-27

Subho Majumdar,
Assistant Professor, Decision Sciences
IIM Bangalore

Session 1
09/09/2026

Schedule

Session	Time	Day	Date
1	02:00-03:30	Wed	09-Sep
2	02:00-03:30	Thu	10-Sep
3	02:00-03:30	Thu	17-Sep
4	02:00-03:30	Wed	23-Sep
5	02:00-03:30	Thu	24-Sep
6	02:00-03:30	Wed	30-Sep
7	02:00-03:30	Thu	01-Oct
8	02:00-03:30	Wed	07-Oct
9	02:00-03:30	Thu	08-Oct
10	02:00-03:30	Wed	14-Oct
11	02:00-03:30	Thu	15-Oct
	October 20-24		Midterm week
12	02:00-03:30	Wed	28-Oct
13	02:00-03:30	Thu	29-Oct
14	02:00-03:30	Wed	04-Nov
15	02:00-03:30	Thu	05-Nov
16	02:00-03:30	Wed	11-Nov
17	02:00-03:30	Thu	12-Nov
18	02:00-03:30	Wed	18-Nov
19	02:00-03:30	Thu	19-Nov
20	02:00-03:30	Wed	25-Nov
	December 07-11		End Term Week

Notes

- No class on Wed, 16 Sep
- No classes on Midterm week
- Last class on Wed, 25 Nov

Evaluation components

- **Assignments (20%)**
 - 4 assignments, hard-copy submission
 - Open-everything
- **Paper reading + presentation (20%)**
 - 6 groups of 2 that I'll assign randomly
 - Pick a paper, read it, present
 - Graded on clarity of presentation (12 min + 3 min Q&A)
- **Term project + presentation (30%)**
 - 4 groups of 3 that I'll assign randomly
 - Project reports to be submitted (hard or soft copy)
 - Graded on statistical rigour in report and clarity of presentation (15 min + 5 min Q&A)
- **Endterm (30%)**
 - Pen-and-paper
 - Open-note, closed-everything-else (including AI)

Timeline

Session	Time	Day	Date	Event
1	02:00-03:30	Wed	09-Sep	
2	02:00-03:30	Thu	10-Sep	
3	02:00-03:30	Thu	17-Sep	
4	02:00-03:30	Wed	23-Sep	Reading paper choices due
5	02:00-03:30	Thu	24-Sep	Assignment 1
6	02:00-03:30	Wed	30-Sep	
7	02:00-03:30	Thu	01-Oct	Term projects given
8	02:00-03:30	Wed	07-Oct	
9	02:00-03:30	Thu	08-Oct	Assignment 2
10	02:00-03:30	Wed	14-Oct	
11	02:00-03:30	Thu	15-Oct	Paper-reading presentation
	October 20-24	Midterm week		
	02:00-03:30	Wed	28-Oct	
13	02:00-03:30	Thu	29-Oct	
14	02:00-03:30	Wed	04-Nov	
15	02:00-03:30	Thu	05-Nov	Assignment 3
16	02:00-03:30	Wed	11-Nov	
17	02:00-03:30	Thu	12-Nov	
18	02:00-03:30	Wed	18-Nov	
19	02:00-03:30	Thu	19-Nov	Assignment 4
20	02:00-03:30	Wed	25-Nov	Term project submission+presentation
	December 07-11	End Term Week		Pen-and-paper endterm

Assignments are given out on Thursdays, due start of class the following Wednesday

Course outline

Session	Material covered
1	Distributions derived from the Normal distribution: Chi-Squared, t, and F Distributions. Distribution of the Sample Mean and Sample Variance.
2-3	General concepts: unbiasedness, consistency, order statistics, quantiles, confidence intervals
4-5	Theory of estimation: parameter estimates, Method of Moments, Maximum Likelihood Estimation, efficiency and the Cramer-Rao Lower Bound, sufficiency, completeness
6-7	Hypothesis testing fundamentals: the Neyman-Pearson paradigm; MP, UMP and UMPU tests; Generalized Likelihood Ratio Tests
8-9	One-sample and Two-sample tests: inference of mean, proportion and variance; comparing independent and paired samples
10	Analysis of Variance (ANOVA): one-way and two-way ANOVA
12-13	Categorical data analysis: fisher's exact test, chi-square test of homogeneity/independence, goodness of fit tests
14-15	Linear regression: assumptions, inference, and diagnostics
16	Experimental Design
17	Bayesian hypothesis testing
18-19	Modern inferential tools: multiple testing, conformal prediction, inference in ML/AI